ABB CASE

Watt the Sigma

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THE PROBLEM

1. LIMITED GRID CAPACITY
Growing demand from AI, data centers and EVs puts pressure on existing electricity infrastructure.

2. HIGH PEAK DEMAND
Simultaneous fast charging can push charging sites beyond their available electrical capacity.

3. SLOWING EV ADOPTION
Lower EV adoption growth creates uncertainty for charging infrastructure investment.

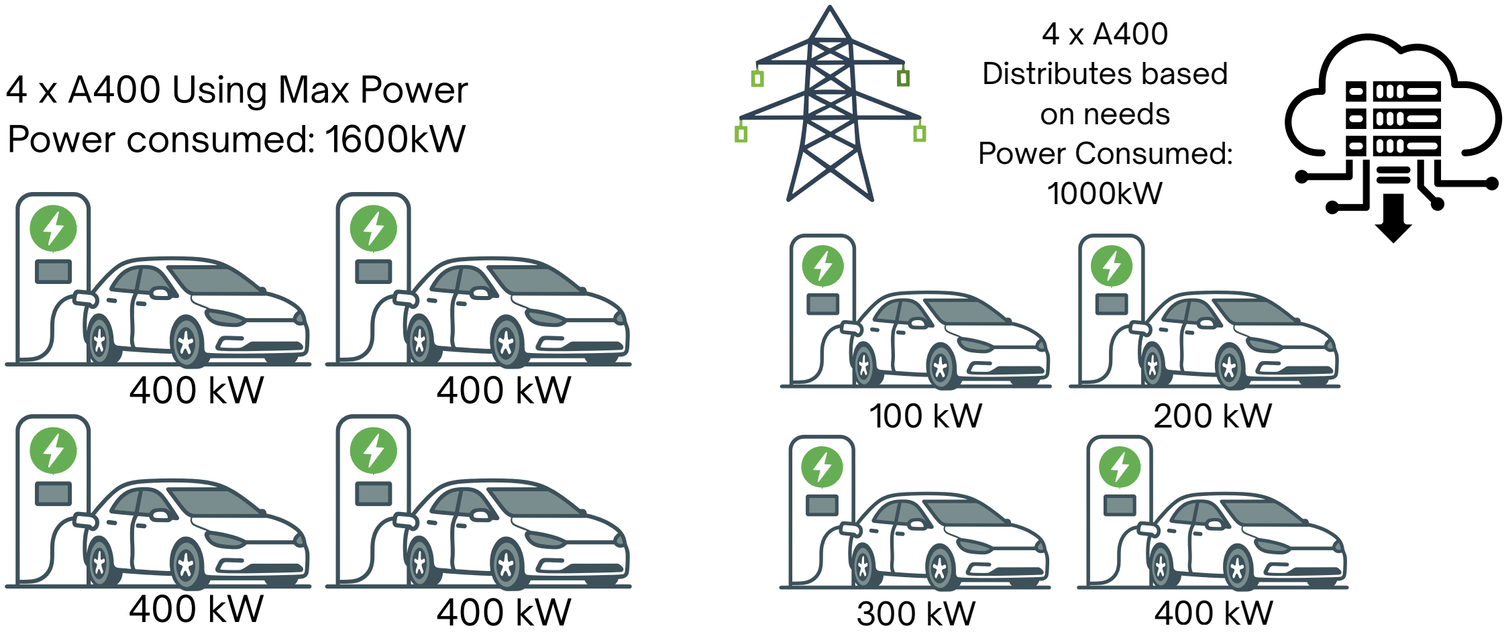
4. FRAGMENTED GRID SYSTEMS
Different provincial and territorial electricity systems make coordination more complex.

SOLUTION Implements usage of products already created

MONITOR — Expanded Energy Monitoring: Develop an integration that collects available electricity usage and capacity data from participating residential and commercial sites within a 25-mile radius, using ReliaHome and compatible energy-management systems.

OPTIMIZE — Centralized Power Management: Send aggregated demand information to OPTIMAX Cloud to help forecast local demand and coordinate EV charging power.

DISTRIBUTE — Priority-Based Charging: Adjust power across connected EV chargers based on available site capacity and customer charging needs.

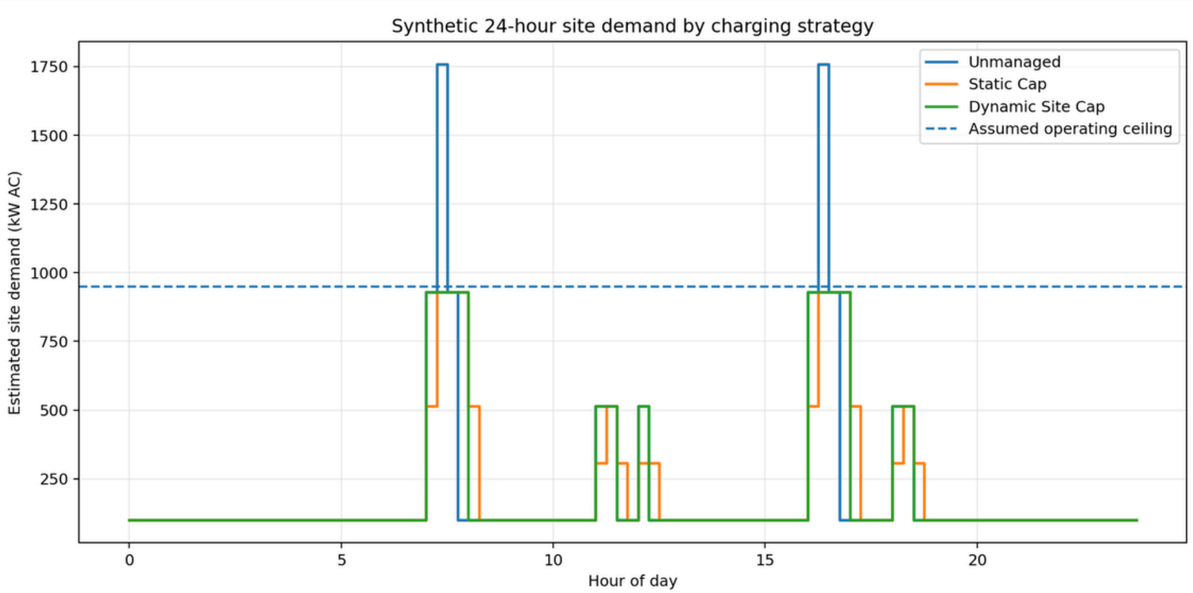


Distribute power through priority-based dynamic load management

SIMULATING SITE-WIDE POWER MANAGEMENT

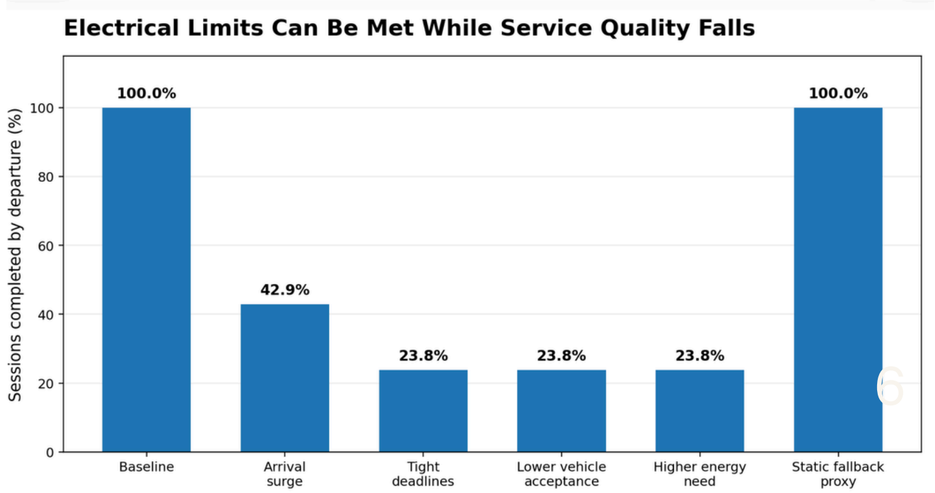
Model inputs

- Four ABB A400 chargers, with two outlets each.
- Assumed site connection limit: 1,000 kW AC.
- Assumed other site load: 100 kW.
- Operating ceiling: 950 kW AC, leaving a 50 kW margin.
- Charger efficiency assumption: 96.5%.



Baseline result

Unmanaged peak	Capped peak
1,758 kW	**929 kW**
Baseline energy	Capped intervals over ceiling
2,100 kWh	**0**

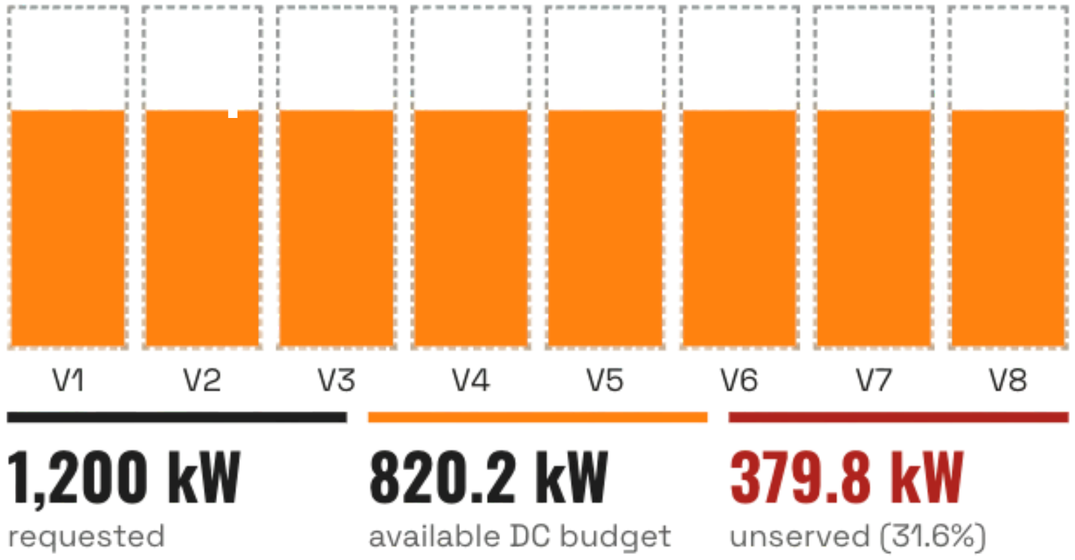


The simulated power cap keeps estimated demand below the assumed 950 kW operating ceiling while preserving baseline energy delivery.

DEMONSTRATION

Overload: demand exceeds the budget

8 vehicles requesting 150 kW each (dashed).
Orange = equal share, 102.5 kW.



Sensitivity to site conditions

DC budget and outlets supportable at 100 kW each. Columns: connection limit (AC). Rows: other load. Outlined = base case.

	800 kW	1,000 kW	1,200 kW
50 kW	**676 kW** 6 of 8 outlets	**868 kW** all 8 outlets	**1,062 kW** all 8 outlets
100 kW	**627 kW** 6 of 8 outlets	**820 kW** all 8 outlets	**1,013 kW** all 8 outlets
200 kW	**531 kW** 5 of 8 outlets	**724 kW** 7 of 8 outlets	**917 kW** all 8 outlets
300 kW	**434 kW** 4 of 8 outlets	**627 kW** 6 of 8 outlets	**820 kW** all 8 outlets

Sources: ABB A400 specifications | ABB Ability OPTIMAX
Github: https://github.com/chichi4nwosu/Watt-the-Sigma

LAUNCH IN QUÉBEC WITH SAFETY AND CUSTOMER TRUST BUILT INTO THE SYSTEM

Dedicated public charging stations first. Local control protects the site; transparent service protects the driver.



PROTECT THE STATION

- Enforce the total site power limit locally.
- Continue safe local control if the cloud disconnects.
- If local control or measurement fails, use verified fallback limits or pause charging.



PROTECT THE DRIVER

- Explain expected charging speed and completion time.
- Use transparent power allocation and prevent indefinite deprioritization.
- Minimize personal-data collection and complete the applicable privacy review.



MAKE DEPLOYMENT PRACTICAL

- Confirm the operator’s electricity contract and pricing plan.
- Verify equipment compatibility and required electrical work.
- Assign installation, commissioning, and support responsibilities before launch.

Operator provides site data → Qualified electrical team verifies installation → ABB integrates and tests → Operator accepts pilot

SCALE ONLY WHEN POWER MANAGEMENT PRESERVES USEFUL CHARGING SERVICE

Evaluate electrical performance, driver outcomes, and operator value together.

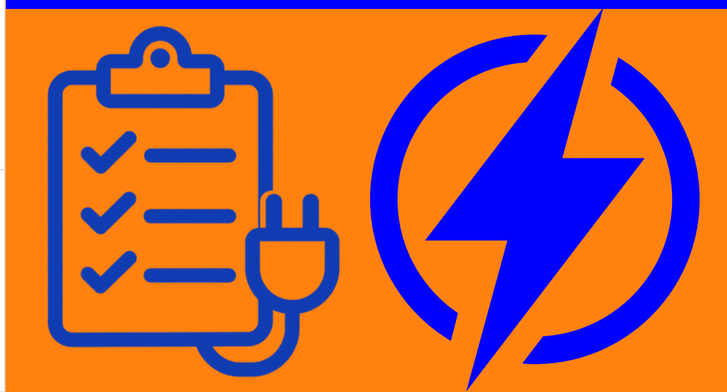
What could undermine the launch?	Our response and proof required
Slower charging drives customers away	Track total time to finish, completed sessions, and abandonment. Adjust allocation if service deteriorates.
Savings disappear under the actual bill	Compare equivalent service using the applicable pricing plan; include installation, fees, and operating costs.
Controls fail or chargers restart together	Test cloud loss, missing measurements, fallback operation, and controlled recovery.
Sustainability claims exceed the evidence	Report measured power and energy separately. Claim emissions reductions or deferred upgrades only with supporting analysis.

PILOT DECISION

Expand only after demonstrating:

- Safe operation under normal and tested failure conditions.
- Acceptable customer service against agreed targets.
- Positive operator value after added costs.
- Repeatable results under representative demand.

If a site fails these tests: revise the controls, narrow the site criteria, or defer rollout.

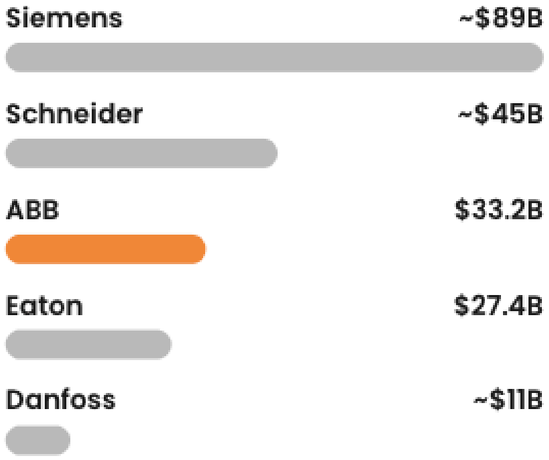


ADOPTION PLAN + COMPETITIVE POSITIONING

For Operators: Offer a 25% discount Year 1 to drive adoption, send operators savings figures and cost-benefit analysis to entice renewal of service.

For Homeowners: Offer 1 free priority charge per month in order to encourage users to adhere to our initiatives charging policies.

2025 REVENUE SCALE (US$B, APPROX.)



Low-voltage drives market: $19.2B in 2023, growing ~4.3% a year to 2032. Also in the field: Yaskawa, Fuji Electric, Rockwell, WEG.

COMPANY	COMPETITIVE ADVANTAGE	PROOF POINT / WATCH-OUT
ABB $33.2B · Motion ~$10B	Motors, drives and electrification in one portfolio	Record $25.3B backlog; Motion margin 19.6%
Schneider €40B+ · +9% organic	EcoStruxure plus AVEVA software; strong data-center pull	Software-led energy management; 18.7% EBITA margin
Siemens €78.9B · +5% comparable	Automation plus digital-twin software; €5.3B software ARR	Data-center revenue €2.9B, up 40%
Eaton $27.4B · +10%	Power management leader, strongest in North America	Electrical backlog up 29% year over year
Danfoss €9.4B · +3% organic	Drives and climate specialist; R&D 5.3% of sales	Direct drives rival in HVAC, water and data centers

FINANCIAL MODEL + ASSUMPTIONS (BASE)

- CAD; $2M startup investment
- $424k annual operating costs
- Aggressive Year 5 targets: 500 operators; 100,000 homeowners
- Home launch: Year 3
- Two paid $3 priority requests/month

ABB Financial Model - 5-year Outlook for Quebec Rollout

Scenario	Base	Base	Base	Base	Base
Cash break-even	Year 4				

Financial plan	Units	Year 1	Year 2	Year 3	Year 4	Year 5
Paying operators	Annual Average	20	80	180	320	500
Annual fee per operator	CAD / operator	$1,800	$2,400	$2,400	$2,400	$2,400
Operator revenue	**CAD**	**$36,000**	**$192,000**	**$432,000**	**$768,000**	**$1,200,000**
Paying Homeowners w/EV	Annual Average	-	-	10,000	40,000	100,000
Annual Fee per Homeowner w/EV	CAD / driver	-	-	$72	$72	$72
Homeowner w/EV revenue	**CAD**	**-**	**-**	**$720,000**	**$2,880,000**	**$7,200,000**
Total revenue	**CAD**	**$36,000**	**$192,000**	**$1,152,000**	**$3,648,000**	**$8,400,000**
Annual operating costs	CAD	$424,000	$424,000	$424,000	$424,000	$424,000
Annual operating profit	**CAD**	**($388,000)**	**($232,000)**	**$728,000**	**$3,224,000**	**$7,976,000**
Cumulative cash after startup cost	**CAD**	**($2,388,000)**	**($2,620,000)**	**($1,892,000)**	**$1,332,000**	**$9,308,000**
Cumulative ROI	***%***	***(119.4%)***	***(131.0%)***	***(94.6%)***	***66.6%***	***465.4%***

FINANCIAL MODEL + ASSUMPTIONS (DOWNSIDE)

- Same startup and operating costs
- Year 5 targets: 150 operators; 25,000 homeowners
- Operator launch: Year 2
- Home launch: Year 4
- One paid $3 priority request/month

ABB Financial Model - 5-year Outlook for Quebec Rollout

Scenario	Downside	Downside	Downside	Downside	Downside
Cash break-even	After Year 5				

Financial plan	Units	Year 1	Year 2	Year 3	Year 4	Year 5
Paying operators	Annual Average	-	10	40	90	150
Annual fee per operator	CAD / operator	-	$2,400	$2,400	$2,400	$2,400
Operator revenue	**CAD**	**-**	**$24,000**	**$96,000**	**$216,000**	**$360,000**
Paying Homeowners w/EV	Annual Average	-	-	-	5,000	25,000
Annual Fee per Homeowner w/EV	CAD / driver	-	-	-	$36	$36
Homeowner w/EV revenue	**CAD**	**-**	**-**	**-**	**$180,000**	**$900,000**
Total revenue	**CAD**	**-**	**$24,000**	**$96,000**	**$396,000**	**$1,260,000**
Annual operating costs	CAD	$424,000	$424,000	$424,000	$424,000	$424,000
Annual operating profit	**CAD**	**($424,000)**	**($400,000)**	**($328,000)**	**($28,000)**	**$836,000**
Cumulative cash after startup cost	**CAD**	**($2,424,000)**	**($2,824,000)**	**($3,152,000)**	**($3,180,000)**	**($2,344,000)**
Cumulative ROI	***%***	***(121.2%)***	***(141.2%)***	***(157.6%)***	***(159.0%)***	***(117.2%)***

*Actual Break Even: Year 7 (assuming continued growth and flat operating costs)

APPENDIX & REFERENCES

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THE END

Thank You!